

Labwork 3: CPU vs GPU Grayscale Conversion

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Code

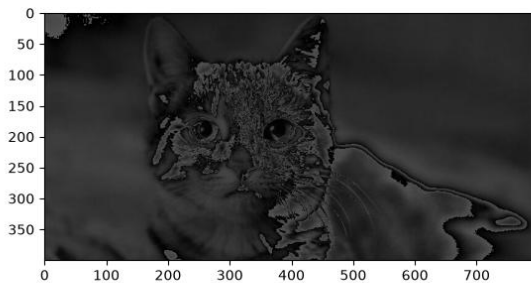
```
blockSizeValues = [32, 64, 128, 256, 512]
img = plt.imread("cat.jpg")
pixelCount = img.shape[0] * img.shape[1]
img_flat = img.reshape(pixelCount, 3)

def grayscale_cpu(src, dst):
    for i in range(src.shape[0]):
        g = np.uint8((src[i, 0] + src[i, 1] + src[i, 2]) / 3)
        dst[i] = g

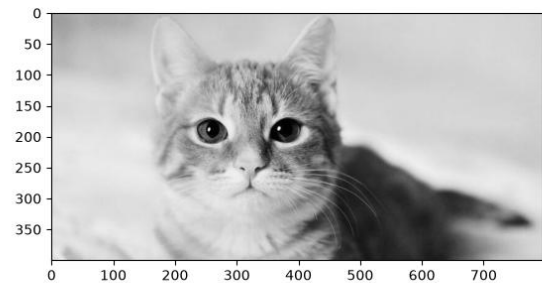
@cuda.jit
def grayscale_gpu(src, dst):
    tid = cuda.threadIdx.x + cuda.blockIdx.x * cuda.blockDim.x
    g = np.uint8((src[tid, 0] + src[tid, 1] + src[tid, 2]) / 3)
    dst[tid, 0] = dst[tid, 1] = dst[tid, 2] = g

devInput = cuda.to_device(img_flat)
devOutput = cuda.device_array((pixelCount, 3), dtype=np.uint8)
grayscale_gpu[gridSize, blockSize](devInput, devOutput)
hostOutput = devOutput.copy_to_host()
```

Block size 64 output



CPU



GPU

Mean time at block size 64 (9 runs): CPU 771.277 ms GPU 1.834 ms

Mean time by block sizes

The performance of GPU is far better than CPU. The CPU performance is not affected by block size. The block size affects GPU performance heavily at 32, and does not change much at 64, 128, 256, and 512. For GPU, the best block size is 128, which gives the lowest mean time of 1.439 ms.

Table 1: Average CPU and GPU processing times by block size

Block size	CPU (ms)	GPU (ms)
32	796.437	151.725
64	771.277	1.834
128	773.547	1.439
256	773.504	1.536
512	779.308	1.499

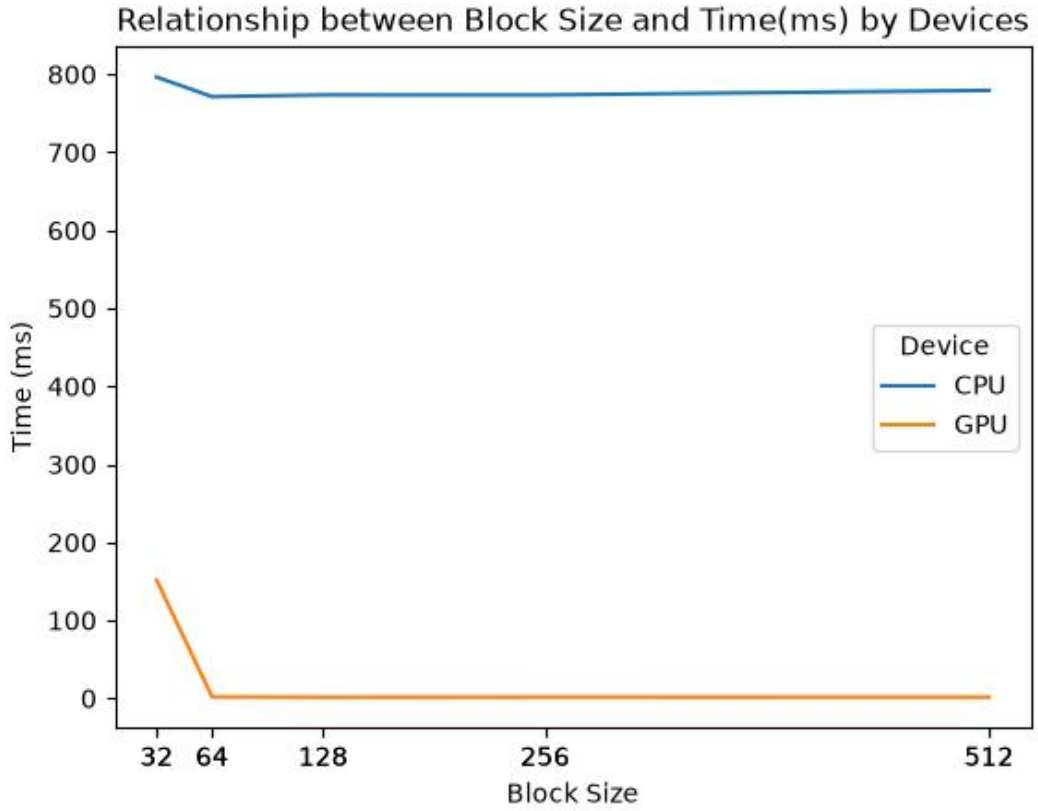


Figure 1: Mean time by block size and device.